ExamGOAL
PYQ PDF

Free · With answer key

JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Differentiation

JEE Main Mathematics · Previous Year Questions with Answer Key

88	79	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	75 questions
Numerical	13 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 75 questions

Q1. JEE Main 2026 (Online) 4th April Morning Shift

If $y = \tan^{-1}\left(\frac{3\cos x - 4\sin x}{4\cos x + 3\sin x}\right) + 2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{3}}{2}$ is equal to :

(A) 3 (B) -1 (C) 1 (D) 2

Open solution

Q2. JEE Main 2026 (Online) 21st January Evening Shift

Let $f(x) = x^3 + x^2 f'(1) + 2x f''(2) + f'''(3), x \in \mathbb{R}$. Then the value of $f'(5)$ is :

(A) $\frac{657}{5}$ (B) $\frac{117}{5}$ (C) $\frac{2}{5}$ (D) $\frac{62}{5}$

Open solution

Q3. JEE Main 2025 (Online) 3rd April Morning Shift

If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}, x \in \mathbb{R}$, then $\frac{d^2y}{dx^2} + y$ is equal to

(A) 28 (B) 27 (C) -1 (D) 1

Open solution

Q4. JEE Main 2025 (Online) 2nd April Morning Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that $(\sin x \cos y)(f(2x+2y) - f(2x-2y)) = (\cos x \sin y)(f(2x+2y) + f(2x-2y))$, for all $x, y \in \mathbf{R}$. If $f'(0) = \frac{1}{2}$, then the value of $24 f''\left(\frac{5\pi}{3}\right)$ is :

(A) 2 (B) 3 (C) -3 (D) -2

Open solution

Q5. JEE Main 2025 (Online) 24th January Evening Shift

Let $f : (0, \infty) \to \mathbf{R}$ be a function which is differentiable at all points of its domain and satisfies the condition $x^2 f'(x) = 2x f(x) + 3$, with $f(1) = 4$. Then $2f(2)$ is equal to :

(A) 19 (B) 23 (C) 29 (D) 39

Open solution

Q6. JEE Main 2024 (Online) 9th April Evening Shift

If $\log_e y = 3\sin^{-1} x$, then $(1-x^2)y'' - xy'$ at $x = \frac{1}{2}$ is equal to

(A) $9e^{\pi/2}$ (B) $9e^{\pi/6}$ (C) $3e^{\pi/2}$ (D) $3e^{\pi/6}$

Open solution

Q7. JEE Main 2024 (Online) 9th April Morning Shift

Let $f(x) = ax^3 + bx^2 + cx + 41$ be such that $f(1) = 40, f'(1) = 2$ and $f''(1) = 4$. Then $a^2 + b^2 + c^2$ is equal to:

(A) 54 (B) 51 (C) 73 (D) 62

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q8. JEE Main 2024 (Online) 6th April Evening Shift

Suppose for a differentiable function $h, h(0)=0, h(1)=1$ and $h'(0)=h'(1)=2$. If $g(x)=h\left(\mathrm{e}^{x}\right)\mathrm{e}^{h(x)}$, then $g'(0)$ is equal to:

(A) 4 (B) 5 (C) 3 (D) 8

Open solution

Q9. JEE Main 2024 (Online) 6th April Morning Shift

If $f(x)=\begin{cases}x^3\sin\left(\frac{1}{x}\right), & x\neq 0\\ 0 & ,x=0\end{cases}$, then

(A) $f''(0)=0$ (B) $f''(0)=1$

(C) $f''\left(\frac{2}{\pi}\right)=\frac{24-\pi^2}{2\pi}$ (D) $f''\left(\frac{2}{\pi}\right)=\frac{12-\pi^2}{2\pi}$

Open solution

Q10. JEE Main 2024 (Online) 6th April Morning Shift

Let $f:(-\infty,\infty)-\{0\}\to\mathbb{R}$ be a differentiable function such that $f'(1)=\lim\limits_{a\to\infty}a^2 f\left(\frac{1}{a}\right)$. Then $\lim\limits_{a\to\infty}\frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right)+a^2-2\log_e a$ is equal to

(A) $\frac{5}{2}+\frac{\pi}{8}$ (B) $\frac{3}{8}+\frac{\pi}{4}$ (C) $\frac{3}{4}+\frac{\pi}{8}$ (D) $\frac{3}{2}+\frac{\pi}{4}$

Open solution

Q11. JEE Main 2024 (Online) 5th April Evening Shift

If $y(\theta)=\frac{2\cos\theta+\cos 2\theta}{\cos 3\theta+4\cos 2\theta+5\cos\theta+2}$, then at $\theta=\frac{\pi}{2}$, $y''+y'+y$ is equal to :

(A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

Open solution

Q12. JEE Main 2024 (Online) 4th April Morning Shift

Let $f(x)=x^5+2\mathrm{e}^{x/4}$ for all $x\in\mathbf{R}$. Consider a function $g(x)$ such that $(g\circ f)(x)=x$ for all $x\in\mathbf{R}$. Then the value of $8g'(2)$ is :

(A) 4 (B) 2 (C) 16 (D) 8

Open solution

Q13. JEE Main 2024 (Online) 31st January Morning Shift

If $f(x)=\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-2\end{vmatrix}$ for all $x\in\mathbb{R}$, then $2f(0)+f'(0)$ is equal to

(A) 24 (B) 18 (C) 42 (D) 48

Open solution

Q14. JEE Main 2024 (Online) 30th January Evening Shift

Let $f:\mathbb{R}-\{0\}\to\mathbb{R}$ be a function satisfying $f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}$ for all $x, y, f(y)\neq 0$. If $f'(1)=2024$, then

(A) $xf'(x)+2024f(x)=0$ (B) $xf'(x)-2023f(x)=0$

(C) $xf'(x)-2024f(x)=0$ (D) $xf'(x)+f(x)=2024$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q15. JEE Main 2024 (Online) 30th January Morning Shift

Let $g : \mathbf{R} \to \mathbf{R}$ be a non constant twice differentiable function such that $g'\left(\frac{1}{2}\right) = g'\left(\frac{3}{2}\right)$. If a real valued function f is defined as $f(x) = \frac{1}{2}[g(x) + g(2-x)]$, then

(A) $f''(x) = 0$ for at least two x in $(0, 2)$

(B) $f'\left(\frac{3}{2}\right) + f'\left(\frac{1}{2}\right) = 1$

(C) $f''(x) = 0$ for no x in $(0, 1)$

(D) $f''(x) = 0$ for exactly one x in $(0, 1)$

Open solution

Q16. JEE Main 2024 (Online) 30th January Morning Shift

If $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$, then $\frac{1}{5}f'(0)$ is equal to :

(A) 2 (B) 1 (C) 0 (D) 6

Open solution

Q17. JEE Main 2024 (Online) 29th January Evening Shift

Let $y = \log_e\left(\frac{1-x^2}{1+x^2}\right), -1 < x < 1$. Then at $x = \frac{1}{2}$, the value of $225\,(y' - y'')$ is equal to

(A) 732 (B) 736 (C) 742 (D) 746

Open solution

Q18. JEE Main 2024 (Online) 29th January Morning Shift

Suppose $f(x) = \frac{(2^x+2^{-x})\tan x\sqrt{\tan^{-1}(x^2-x+1)}}{(7x^2+3x+1)^3}$. Then the value of $f'(0)$ is equal to

(A) π (B) $\sqrt{\pi}$ (C) 0 (D) $\frac{\pi}{2}$

Open solution

Q19. JEE Main 2023 (Online) 13th April Morning Shift

For the differentiable function $f : \mathbb{R} - \{0\} \to \mathbb{R}$, let $3f(x) + 2f\left(\frac{1}{x}\right) = \frac{1}{x} - 10$, then $\left|f(3) + f'\left(\frac{1}{4}\right)\right|$ is equal to

(A) 13 (B) $\frac{29}{5}$ (C) $\frac{33}{5}$ (D) 7

Open solution

Q20. JEE Main 2023 (Online) 8th April Morning Shift

Let $f(x) = \frac{\sin x+\cos x-\sqrt{2}}{\sin x-\cos x}, x \in [0, \pi] - \left\{\frac{\pi}{4}\right\}$. Then $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ is equal to

(A) $\frac{2}{3\sqrt{3}}$ (B) $\frac{2}{9}$ (C) $\frac{-1}{3\sqrt{3}}$ (D) $\frac{-2}{3}$

Open solution

Q21. JEE Main 2023 (Online) 6th April Morning Shift

If $2x^y + 3y^x = 20$, then $\frac{dy}{dx}$ at $(2, 2)$ is equal to :

(A) $-\left(\frac{3+\log_e 16}{4+\log_e 8}\right)$ (B) $-\left(\frac{2+\log_e 8}{3+\log_e 4}\right)$

(C) $-\left(\frac{3+\log_e 8}{2+\log_e 4}\right)$ (D) $-\left(\frac{3+\log_e 4}{2+\log_e 8}\right)$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q22. JEE Main 2023 (Online) 1st February Evening Shift

If $y(x) = x^x, x > 0$, then $y''(2) - 2y'(2)$ is equal to

(A) $4(\log_e 2)^2 + 2$ (B) $8\log_e 2 - 2$ (C) $4\log_e 2 + 2$ (D) $4(\log_e 2)^2 - 2$

Open solution

Q23. JEE Main 2023 (Online) 1st February Morning Shift

Let $f(x) = 2x + \tan^{-1}x$ and $g(x) = \log_e(\sqrt{1+x^2} + x), x \in [0, 3]$. Then

(A) there exists $\widehat{x} \in [0, 3]$ such that $f'(\widehat{x}) < g'(\widehat{x})$

(B) there exist $0 < x_1 < x_2 < 3$ such that $f(x) < g(x), \forall x \in (x_1, x_2)$

(C) $\min f'(x) = 1 + \max g'(x)$

(D) $\max f(x) > \max g(x)$

Open solution

Q24. JEE Main 2023 (Online) 31st January Morning Shift

Let $y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{\frac{3}{2}}\right)\right)\right)$. Then, at x = 1,

(A) $2y' + \sqrt{3}\pi^2 y = 0$ (B) $y' + 3\pi^2 y = 0$

(C) $\sqrt{2}y' - 3\pi^2 y = 0$ (D) $2y' + 3\pi^2 y = 0$

Open solution

Q25. JEE Main 2023 (Online) 29th January Evening Shift

Let f and g be the twice differentiable functions on $\mathbb{R}$ such that

$f''(x) = g''(x) + 6x$

$f'(1) = 4g'(1) - 3 = 9$

$f(2) = 3g(2) = 12.$

Then which of the following is NOT true?

(A) $g(-2) - f(-2) = 20$

(B) There exists $x_0 \in (1, 3/2)$ such that $f(x_0) = g(x_0)$

(C) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

(D) If $-1 < x < 2$, then $|f(x) - g(x)| < 8$

Open solution

Q26. JEE Main 2023 (Online) 25th January Morning Shift

Let $y(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$. Then $y' - y''$ at $x = -1$ is equal to

(A) 496 (B) 976 (C) 464 (D) 944

Open solution

Q27. JEE Main 2023 (Online) 24th January Evening Shift

If $f(x) = x^3 - x^2 f'(1) + xf''(2) - f'''(3), x \in \mathbb{R}$, then

(A) $2f(0) - f(1) + f(3) = f(2)$ (B) $f(1) + f(2) + f(3) = f(0)$

(C) $f(3) - f(2) = f(1)$ (D) $3f(1) + f(2) = f(3)$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q28. JEE Main 2022 (Online) 28th July Evening Shift

Let $x(t) = 2\sqrt{2}\cos t\sqrt{\sin 2t}$ and

$y(t) = 2\sqrt{2}\sin t\sqrt{\sin 2t}, t \in \left(0, \frac{\pi}{2}\right)$.

Then $\frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}$ at $t = \frac{\pi}{4}$ is equal to :

(A) $\frac{-2\sqrt{2}}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{-2}{3}$

Open solution

Q29. JEE Main 2022 (Online) 26th July Evening Shift

The value of $\log_e 2\frac{d}{dx}(\log_{\cos x}\operatorname{cosec} x)$ at $x = \frac{\pi}{4}$ is

(A) $-2\sqrt{2}$ (B) $2\sqrt{2}$ (C) -4 (D) 4

Open solution

Q30. JEE Main 2022 (Online) 27th June Morning Shift

If $\cos^{-1}\left(\frac{y}{2}\right) = \log_e\left(\frac{x}{5}\right)^5$, $|y| < 2$, then :

(A) $x^2y'' + xy' - 25y = 0$ (B) $x^2y'' - xy' - 25y = 0$

(C) $x^2y'' - xy' + 25y = 0$ (D) $x^2y'' + xy' + 25y = 0$

Open solution

Q31. JEE Main 2022 (Online) 25th June Morning Shift

Let f : R $\to$ R be defined as $f(x) = x^3 + x - 5$. If g(x) is a function such that $f(g(x)) = x, \forall' x' \in R$, then g'(63) is equal to _____.

(A) $\frac{1}{49}$ (B) $\frac{3}{49}$ (C) $\frac{43}{49}$ (D) $\frac{91}{49}$

Open solution

Q32. JEE Main 2022 (Online) 24th June Evening Shift

If $y = \tan^{-1}\left(\sec x^3 - \tan x^3\right)$, $\frac{\pi}{2} < x^3 < \frac{3\pi}{2}$, then

(A) $xy'' + 2y' = 0$ (B) $x^2y'' - 6y + \frac{3\pi}{2} = 0$

(C) $x^2y'' - 6y + 3\pi = 0$ (D) $xy'' - 4y' = 0$

Open solution

Q33. JEE Main 2021 (Online) 27th August Evening Shift

If $y(x) = \cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right), x \in \left(\frac{\pi}{2}, \pi\right)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{6}$ is :

(A) $-\frac{1}{2}$ (B) -1 (C) $\frac{1}{2}$ (D) 0

Open solution

Q34. JEE Main 2021 (Online) 26th August Morning Shift

Let $f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right)$, 0 < x < 1. Then :

(A) $(1-x)^2 f'(x) - 2(f(x))^2 = 0$ (B) $(1+x)^2 f'(x) + 2(f(x))^2 = 0$

(C) $(1-x)^2 f'(x) + 2(f(x))^2 = 0$ (D) $(1+x)^2 f'(x) - 2(f(x))^2 = 0$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q35. JEE Main 2020 (Online) 5th September Evening Slot

The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect to $\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ at x = $\frac{1}{2}$ is :

(A) $\frac{2\sqrt{3}}{3}$ (B) $\frac{2\sqrt{3}}{5}$ (C) $\frac{\sqrt{3}}{10}$ (D) $\frac{\sqrt{3}}{12}$

Open solution

Q36. JEE Main 2020 (Online) 4th September Morning Slot

If $\left(a+\sqrt{2}b\cos x\right)\left(a-\sqrt{2}b\cos y\right)=a^2-b^2$

where a > b > 0, then $\frac{dx}{dy}$ at $\left(\frac{\pi}{4},\frac{\pi}{4}\right)$ is :

(A) $\frac{a-2b}{a+2b}$ (B) $\frac{a-b}{a+b}$ (C) $\frac{a+b}{a-b}$ (D) $\frac{2a+b}{2a-b}$

Open solution

Q37. JEE Main 2020 (Online) 3rd September Morning Slot

If $y^2 + \log_e(\cos^2 x) = y$, $x \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, then :

(A) |y"(0)| = 2 (B) |y'(0)| + |y"(0)| = 3

(C) y"(0) = 0 (D) |y'(0)| + |y"(0)| = 1

Open solution

Q38. JEE Main 2020 (Online) 9th January Evening Slot

If $x = 2\sin\theta - \sin 2\theta$ and $y = 2\cos\theta - \cos 2\theta$, $\theta \in [0, 2\pi]$, then $\frac{d^2y}{dx^2}$ at $\theta = \pi$ is :

(A) $\frac{3}{8}$ (B) $\frac{3}{2}$ (C) $\frac{3}{4}$ (D) $-\frac{3}{4}$

Open solution

Q39. JEE Main 2020 (Online) 9th January Evening Slot

Let f and g be differentiable functions on R such that fog is the identity function. If for some a, b $\in$ R, g'(a) = 5 and g(a) = b, then f'(b) is equal to :

(A) 1 (B) 5 (C) $\frac{2}{5}$ (D) $\frac{1}{5}$

Open solution

Q40. JEE Main 2020 (Online) 8th January Morning Slot

Let f(x) = $(\sin(\tan^{-1}x) + \sin(\cot^{-1}x))^2 - 1$, |x| > 1.
If $\frac{dy}{dx} = \frac{1}{2}\frac{d}{dx}\left(\sin^{-1}\left(f\left(x\right)\right)\right)$ and $y\left(\sqrt{3}\right) = \frac{\pi}{6}$, then $y(-\sqrt{3})$ is equal to :

(A) $\frac{5\pi}{6}$ (B) $-\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q41. JEE Main 2020 (Online) 7th January Evening Slot

Let y = y(x) be a function of x satisfying

$y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and

$y\left(\frac{1}{2}\right) = -\frac{1}{4}$. Then $\frac{dy}{dx}$ at x = $\frac{1}{2}$, is equal to :

(A) $\frac{2}{\sqrt{5}}$ (B) $-\frac{\sqrt{5}}{2}$ (C) $\frac{\sqrt{5}}{2}$ (D) $-\frac{\sqrt{5}}{4}$

Open solution

Q42. JEE Main 2020 (Online) 7th January Morning Slot

Let $x^k + y^k = a^k$, (a, k > 0) and $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$, then k is:

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{3}{2}$

Open solution

Q43. JEE Main 2020 (Online) 7th January Morning Slot

If $y(\alpha) = \sqrt{2\left(\frac{\tan\alpha+\cot\alpha}{1+\tan^2\alpha}\right) + \frac{1}{\sin^2\alpha}}, \alpha \in \left(\frac{3\pi}{4}, \pi\right)$

$\frac{dy}{d\alpha}\ at\ \alpha = \frac{5\pi}{6} is$:

(A) 4 (B) -4 (C) $\frac{4}{3}$ (D) $-\frac{1}{4}$

Open solution

Q44. JEE Main 2019 (Online) 12th April Evening Slot

The derivative of $\tan^{-1}\left(\frac{\sin x-\cos x}{\sin x+\cos x}\right)$, with respect to $\frac{x}{2}$, where $\left(x \in \left(0, \frac{\pi}{2}\right)\right)$ is :

(A) 1 (B) 2 (C) $\frac{2}{3}$ (D) $\frac{1}{2}$

Open solution

Q45. JEE Main 2019 (Online) 12th April Morning Slot

If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$ at x = 0 is equal to :

(A) $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$ (B) $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$

(C) $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$ (D) $\left(\frac{1}{e}, \frac{1}{e^2}\right)$

Open solution

Q46. JEE Main 2019 (Online) 10th April Evening Slot

Let f(x) = $\log_e(\sin x)$, (0 < x < π) and g(x) = $\sin^{-1}(e^{-x})$, ($x \geq 0$). If α is a positive real number such that a = (fog)'(α) and b = (fog)(α), then :

(A) $a\alpha^2 + b\alpha - a = -2\alpha^2$ (B) $a\alpha^2 + b\alpha + a = 0$

(C) $a\alpha^2 - b\alpha - a = 0$ (D) $a\alpha^2 - b\alpha - a = 1$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q47. JEE Main 2019 (Online) 8th April Evening Slot

If $f(1) = 1$, $f'(1) = 3$, then the derivative of $f(f(f(x))) + (f(x))^2$ at $x = 1$ is :

(A) 33 (B) 12 (C) 9 (D) 15

Open solution

Q48. JEE Main 2019 (Online) 8th April Morning Slot

If $2y = \left(\cot^{-1}\left(\frac{\sqrt{3}\cos x+\sin x}{\cos x-\sqrt{3}\sin x}\right)\right)^2$,

$x \in \left(0, \frac{\pi}{2}\right)$ then $\frac{dy}{dx}$ is equal to:

(A) $2x - \frac{\pi}{3}$ (B) $\frac{\pi}{6} - x$ (C) $\frac{\pi}{3} - x$ (D) $x - \frac{\pi}{6}$

Open solution

Q49. JEE Main 2019 (Online) 12th January Morning Slot

For $x > 1$, if $(2x)^{2y} = 4e^{2x-2y}$,

then $(1 + \log_e 2x)^2 \frac{dy}{dx}$ is equal to :

(A) $\frac{x\log_e 2x-\log_e 2}{x}$ (B) $\log_e 2x$

(C) $x \log_e 2x$ (D) $\frac{x\log_e 2x+\log_e 2}{x}$

Open solution

Q50. JEE Main 2019 (Online) 11th January Morning Slot

If $x\log_e(\log_e x) - x^2 + y^2 = 4 (y > 0)$, then $\frac{dy}{dx}$ at $x = e$ is equal to :

(A) $\frac{(1+2e)}{2\sqrt{4+e^2}}$ (B) $\frac{(1+2e)}{\sqrt{4+e^2}}$

(C) $\frac{(2e-1)}{2\sqrt{4+e^2}}$ (D) $\frac{e}{\sqrt{4+e^2}}$

Open solution

Q51. JEE Main 2019 (Online) 10th January Morning Slot

Let $f : R \to R$ be a function such that $f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3)$, $x \in R$. Then $f(2)$ equals -

(A) 30 (B) -2 (C) -4 (D) 8

Open solution

Q52. JEE Main 2019 (Online) 9th January Evening Slot

If $x = 3\tan t$ and $y = 3\sec t$, then the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$, is :

(A) $\frac{1}{3\sqrt{2}}$ (B) $\frac{1}{6\sqrt{2}}$ (C) $\frac{3}{2\sqrt{2}}$ (D) $\frac{1}{6}$

Open solution

Q53. JEE Main 2018 (Online) 16th April Morning Slot

If $x = \sqrt{2^{\cos ec^{-1}t}}$ and $y = \sqrt{2^{sec^{-1}t}}$ $(|t| \geq 1)$, then $\frac{dy}{dx}$ is equal to :

(A) $\frac{y}{x}$ (B) $\frac{x}{y}$ (C) $-\frac{y}{x}$ (D) $-\frac{x}{y}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q54. JEE Main 2018 (Online) 15th April Evening Slot

If $f(x) = \sin^{-1}\left(\frac{2\times 3^x}{1+9^x}\right)$, then $f'\left(-\frac{1}{2}\right)$ equals :

(A) $-\sqrt{3}\log_e\sqrt{3}$ (B) $\sqrt{3}\log_e\sqrt{3}$ (C) $-\sqrt{3}\log_e 3$ (D) $\sqrt{3}\log_e 3$

Open solution

Q55. JEE Main 2018 (Online) 15th April Morning Slot

If $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$, then $\lim_{x\to 0}\frac{f'(x)}{x}$

(A) does not exist. (B) exists and is equal to 2.

(C) exists and is equal to 0. (D) exists and is equal to -2.

Open solution

Q56. JEE Main 2018 (Online) 15th April Morning Slot

If $x^2 + y^2 + \sin y = 4$, then the value of $\frac{d^2y}{dx^2}$ at the point $(-2,0)$ is :

(A) -34 (B) -32 (C) 4 (D) -2

Open solution

Q57. JEE Main 2017 (Online) 9th April Morning Slot

Let f be a polynomial function such that

$f(3x) = f'(x) \cdot f''(x)$, for all $x \in \mathbf{R}$. Then :

(A) $f(2) + f'(2) = 28$ (B) $f''(2) - f'(2) = 0$

(C) $f''(2) - f(2) = 4$ (D) $f(2) - f'(2) + f''(2) = 10$

Open solution

Q58. JEE Main 2017 (Online) 8th April Morning Slot

If $y = \left[x + \sqrt{x^2-1}\right]^{15} + \left[x - \sqrt{x^2-1}\right]^{15}$,

then $(x^2-1)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is equal to :

(A) 125 y (B) 124 y^2 (C) 225 y^2 (D) 225 y

Open solution

Q59. JEE Main 2017 (Offline)

If for $x \in \left(0, \frac{1}{4}\right)$, the derivatives of

$\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ is $\sqrt{x}.g(x)$, then $g(x)$ equals

(A) $\frac{3x\sqrt{x}}{1-9x^3}$ (B) $\frac{3x}{1-9x^3}$

(C) $\frac{3}{1+9x^3}$ (D) $\frac{9}{1+9x^3}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q60. JEE Main 2014 (Online) 12th April Morning Slot

If f(x) = $x^2 - x + 5$, $x > \frac{1}{2}$, and g(x) is its inverse function, then g'(7) equals :

(A) $-\frac{1}{3}$ (B) $\frac{1}{13}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{13}$

Open solution

Q61. JEE Main 2014 (Online) 9th April Morning Slot

If $y = e^{nx}$, then $\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right)$ is equal to :

(A) $n\,e^{nx}$ (B) $n\,e^{-nx}$ (C) 1 (D) $-n\,e^{-nx}$

Open solution

Q62. JEE Main 2014 (Offline)

If g is the inverse of a function f and $f'(x) = \frac{1}{1+x^5}$, then $g'(x)$ is equal to:

(A) $\frac{1}{1+\{g(x)\}^5}$ (B) $1+\{g(x)\}^5$

(C) $1+x^5$ (D) $5x^4$

Open solution

Q63. JEE Main 2013 (Online) 23th April Morning Slot

If $f(x) = \sin(\sin x)$ and $f''(x) + \tan x f'(x) + g(x) = 0$, then $g(x)$ is

(A) $\cos^2 x \cos(\sin x)$ (B) $\sin^2 x \cos(\cos x)$ (C) $\sin^2 x \sin(\cos x)$ (D) $\cos^2 x \sin(\sin x)$

Open solution

Q64. JEE Main 2013 (Online) 22th April Morning Slot

For a > 0, t $\in (0, \frac{\pi}{2})$, let x = $\sqrt{a^{\sin^{-1}t}}$ and y = $\sqrt{a^{\cos^{-1}t}}$, then, $1+\left(\frac{dy}{dx}\right)^2$ equals :

(A) $\frac{x^2}{y^2}$ (B) $\frac{y^2}{x^2}$ (C) $\frac{x^2+y^2}{y^2}$ (D) $\frac{x^2+y^2}{x^2}$

Open solution

Q65. JEE Main 2013 (Online) 9th April Morning Slot

Let $f(x) = \frac{x^2-x}{x^2+2x}$. $x \neq 0, -2$. Then $\frac{d}{dx}\left[f^{-1}(x)\right]$ (wherever it is defined) is equal to:

(A) $\frac{-1}{(1-x)^2}$ (B) $\frac{3}{(1-x)^2}$ (C) $\frac{1}{(1-x)^2}$ (D) $\frac{-3}{(1-x)^2}$

Open solution

Q66. JEE Main 2013 (Offline)

If $y = \sec\left(\tan^{-1}x\right)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) 1 (D) $\sqrt{2}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q67. AIEEE 2012 (Online) 12th May

If $f'(x) = \sin(\log x)$ and $y = f\left(\frac{2x+3}{3-2x}\right)$, then $\frac{dy}{dx}$ equals

(A) $\sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$

(B) $\frac{12}{(3-2x)^2}$

(C) $\frac{12}{(3-2x)^2}\sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$

(D) $\frac{12}{(3-2x)^2}\cos\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$

Open solution

Q68. AIEEE 2011

$\frac{d^2x}{dy^2}$ equals:

(A) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$

(B) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$

(C) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

(D) $\left(\frac{d^2y}{dx^2}\right)^{-1}$

Open solution

Q69. AIEEE 2010

Let $f:(-1,1)\to R$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let $g(x) = [f(2f(x)+2)]^2$. Then $g'(0) =$

(A) -4 (B) 0 (C) -2 (D) 4

Open solution

Q70. AIEEE 2009

Let y be an implicit function of x defined by $x^{2x} - 2x^x \cot y - 1 = 0$. Then $y'(1)$ equals

(A) 1 (B) $\log 2$ (C) $-\log 2$ (D) -1

Open solution

Q71. AIEEE 2006

If $x^m.y^n = (x+y)^{m+n}$, then $\frac{dy}{dx}$ is

(A) $\frac{y}{x}$ (B) $\frac{x+y}{xy}$ (C) xy (D) $\frac{x}{y}$

Open solution

Q72. AIEEE 2004

If $x = e^{y+e^{y+\ldots\ldots\infty}}$, $x>0$, then $\frac{dy}{dx}$ is

(A) $\frac{1+x}{x}$ (B) $\frac{1}{x}$ (C) $\frac{1-x}{x}$ (D) $\frac{x}{1+x}$

Open solution

Q73. AIEEE 2003

If $f(x) = x^n$, then the value of $f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + \ldots\ldots\ldots \frac{(-1)^n f^n(1)}{n!}$ is

(A) 1 (B) 2^n (C) $2^n - 1$ (D) 0

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q74. AIEEE 2003

Let $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in $A.P$, then $f'(a), f'(b), f'(c)$ are in

(A) Arithmetic -Geometric Progression
(B) $A.P$
(C) $G.P$
(D) $H.P$

Open solution

Q75. AIEEE 2002

If $y = \left(x + \sqrt{1+x^2}\right)^n$, then $\left(1+x^2\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is

(A) n^2y
(B) $-n^2y$
(C) $-y$
(D) $2x^2y$

Open solution

Numerical 13 questions

Q76. JEE Main 2025 (Online) 2nd April Morning Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be a thrice differentiable odd function satisfying $f'(x) \geq 0, f''(x) = f(x), f(0) = 0, f'(0) = 3$. Then $9f(\log_e 3)$ is equal to _____ .

Open solution

Q77. JEE Main 2024 (Online) 4th April Evening Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a thrice differentiable function such that $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeros of $(3f'f'' + ff''')(x)$ is _____.

Open solution

Q78. JEE Main 2024 (Online) 1st February Evening Shift

If $y = \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$, then $96y'\left(\frac{\pi}{6}\right)$ is equal to :

Open solution

Q79. JEE Main 2024 (Online) 27th January Morning Shift

Let $f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3), x \in \mathbf{R}$. Then $f'(10)$ is equal to _____.

Open solution

Q80. JEE Main 2023 (Online) 13th April Evening Shift

Let $f(x) = \sum_{k=1}^{10} kx^k, x \in \mathbb{R}$. If $2f(2) + f'(2) = 119(2)^n + 1$ then n is equal to _____

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q81. JEE Main 2023 (Online) 1st February Morning Shift

If $f(x) = x^2 + g'(1)x + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$, then the value of $f(4) - g(4)$ is equal to _____.

Open solution

Q82. JEE Main 2023 (Online) 29th January Morning Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function that satisfies the relation $f(x + y) = f(x) + f(y) - 1, \forall x, y \in \mathbb{R}$. If $f'(0) = 2$, then $|f(-2)|$ is equal to _____.

Open solution

Q83. JEE Main 2022 (Online) 27th July Evening Shift

For the curve $C : (x^2 + y^2 - 3) + (x^2 - y^2 - 1)^5 = 0$, the value of $3y' - y^3y''$, at the point (α, α), $\alpha > 0$, on C, is equal to _____.

Open solution

Q84. JEE Main 2022 (Online) 27th June Evening Shift

If $y(x) = (x^x)^x$, $x > 0$, then $\frac{d^2x}{dy^2} + 20$ at x = 1 is equal to _____.

Open solution

Q85. JEE Main 2022 (Online) 26th June Evening Shift

Let f : R $\to$ R satisfy $f(x + y) = 2^x f(y) + 4^y f(x)$, $\forall$x, y $\in$ R. If f(2) = 3, then $14. \frac{f'(4)}{f'(2)}$ is equal to _____.

Open solution

Q86. JEE Main 2021 (Online) 26th August Morning Shift

If y = y(x) is an implicit function of x such that $\log_e(x + y) = 4xy$, then $\frac{d^2y}{dx^2}$ at x = 0 is equal to _____.

Open solution

Q87. JEE Main 2021 (Online) 17th March Morning Shift

If $f(x) = \sin\left(\cos^{-1}\left(\frac{1-2^{2x}}{1+2^{2x}}\right)\right)$ and its first derivative with respect to x is $-\frac{b}{a}\log_e 2$ when x = 1, where a and b are integers, then the minimum value of $|a^2 - b^2|$ is _____ .

Open solution

Q88. JEE Main 2020 (Online) 2nd September Evening Slot

If $y = \sum_{k=1}^{6} k\cos^{-1}\left\{\frac{3}{5}\cos kx - \frac{4}{5}\sin kx\right\}$,

then $\frac{dy}{dx}$ at x = 0 is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **B**	3. **C**	4. **C**	5. **D**
6. **A**	7. **B**	8. **A**	9. **C**	10. *Scan QR*
11. **D**	12. **C**	13. **C**	14. **C**	15. **A**
16. **C**	17. **B**	18. **B**	19. **A**	20. **B**
21. **B**	22. **D**	23. **D**	24. **D**	25. **D**
26. **A**	27. **A**	28. **D**	29. **D**	30. **D**
31. **A**	32. **B**	33. **A**	34. **C**	35. **C**
36. **C**	37. **A**	38. **A**	39. **D**	40. **B**
41. **B**	42. **B**	43. **A**	44. **B**	45. **B**
46. **D**	47. **A**	48. **D**	49. **A**	50. **C**
51. **B**	52. **B**	53. **C**	54. **B**	55. **D**
56. **A**	57. **B**	58. **D**	59. **D**	60. **C**
61. **D**	62. **B**	63. **D**	64. **D**	65. **B**
66. **A**	67. **C**	68. **C**	69. **A**	70. **D**
71. **A**	72. **C**	73. **D**	74. **B**	75. **A**

Numerical

76. **36**	77. **5**	78. **105**	79. **202**	80. **10**
81. **14**	82. **3**	83. **16**	84. **16**	85. **248**
86. **40**	87. **481**	88. **91**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs